Inorganic Cumulative Exam
September 6, 2013

1) The complex $[Ni(CN)_4]^{-2}$ is diamagnetic but $NiCl_4^{-2}$ is paramagnetic with 2 unpaired electrons. Likewise $Fe(CN)_6^{-3}$ has only one unpaired electron by $Fe(H_2O)_6^{+3}$ has five. Explain these experimental observations using simple crystal field theory.

2) Diamagnetic complexes of Co(III) such as $Co(NH_3)_6^{3+}$, $Co(en)_3^{+3}$ and $Co(NO_2)_6^{-3}$ are orange-yellow. In contrast, the paramagnetic complexes CoF_6^{-3} and $Co(H_2O)_3F_3$ are blue. Explain qualitatively this difference in color.

3) For which member of the following pairs of complexes would Δ be the larger

$Cr(H_2O)_6^{2+}$, $Cr(H_2O)_6^{3+}$ $\quad$ $Ni(H_2O)_6^{2+}$, $Ni(en)_3^{2+}$

CrF_6^{-3}, $Cr(NH_3)_6^{3+}$ $\quad$ MnF_6^{-2} $\quad$ ReF_6^{-2}

$Fe(CN)_6^{-4}$, $Fe(CN)_6^{-3}$ $\quad$ $Co(en)_3^{3+}$, $\quad$ $Rh(en)_3^{3+}$

4) In each of the following complexes, rationalize the numbers of observed unpaired electrons:

$Ni(CN)_6^{-4}$	1
$Mn(CN)_6^{-2}$	3
$Cr(en)_3^{2+}$	4
$Fe(ox)_3^{-3}$	5
$Pd(CN)_4^{-2}$	0
$CoCl_4^{-2}$	3
$NiBr_4^{-2}$	2

5) What is the ground state term symbol for Mn^{3+}, Co^{2+}, Fe^{2+}, Cr^{3+}